COMP4423 Assignment 2

PolyU Main Campus Image Classification

Name: Huang Haoran

Student number: 24101322D

Course: COMP4423 Computer Vision, 2026/27 Semester 1

Deadline: 23:59 HKT, 28 October 2026

Code: https://github.com/Retr0-rgb-lab/COMP4423_Assignment2

Dataset link (the same string as `DATASET_LINK.txt`):
https://drive.google.com/drive/folders/1_ld7gsHLhzyFJY3bUgTIwoOeVXRDFuAr

The complete photograph set is on Google Drive as 27 zip parts named `COMP4423_A2_dataset_part01.zip` through `COMP4423_A2_dataset_part27.zip`. Download all 27 and unpack them into one directory; the file names inside are the ones used in this report. The link is shared as *anyone with the link*, so no sign-in is needed.

1. Introduction

1.1 Problem and motivation

The program takes one scene photograph and predicts whether the camera was inside the Hong Kong Polytechnic University main campus. The positive class is that main campus only. A building that belongs to PolyU but sits off the main campus is outside. The label is geographic. It is not ``indoors versus outdoors'', and it is not ``red brick versus something else''.

The practical question is whether a traditional classifier, reading pixels alone, can tell those two geographies apart when many scenes look alike: walkways, courtyards, lawns, and red-brick façades exist on both sides of the boundary. The Hong Kong University Main Building was collected for that reason. It is red brick, and it is outside PolyU.

1.2 Class definitions

The two classes are the ones the official campus map defines, with no local reinterpretation.

Label	Definition used in this project
inside (1)	The camera and the main scene are inside the PolyU main campus, using the official campus map as the boundary [10].
outside (0)	The camera and the main scene are outside that campus. Affiliated sites off the main campus stay outside.

Uncertain boundary views are excluded: gate thresholds, covered turnstiles at the entrance, connecting bridges, and shots that look across the boundary. The camera and the subject must be on the same side. Labels are not assigned from red brick, a logo, or architectural style. The classifier receives pixels only. GPS, EXIF, file names, folder names, and map lookups may be used to check a label or to find a file. They are not features.

At the main entrance, 11 photographs were removed under this rule: 4 taken at the campus name board, where the side of the gate is ambiguous, and 7 in the covered passage at the turnstiles, which is a threshold and also contains recognizable people.

1.3 Pipeline in one pass

1. Photographs are grouped by physical place. One place is one `group_id`.
2. Groups are split into train, validation, and test. A group is never split.
3. Each training and evaluation image is resized and center-cropped to 256×256 by `resize_and_crop` in `src/task3/preprocess.py`.
4. HOG and uniform LBP are computed on the grayscale crop, standardized separately, and concatenated.
5. A random forest (300 trees, maximum depth 12) outputs outside or inside.
6. The fitted object is `models/task3_pipeline.joblib`.

Figure 1 shows the same path as a diagram, together with the order in which the model was chosen.

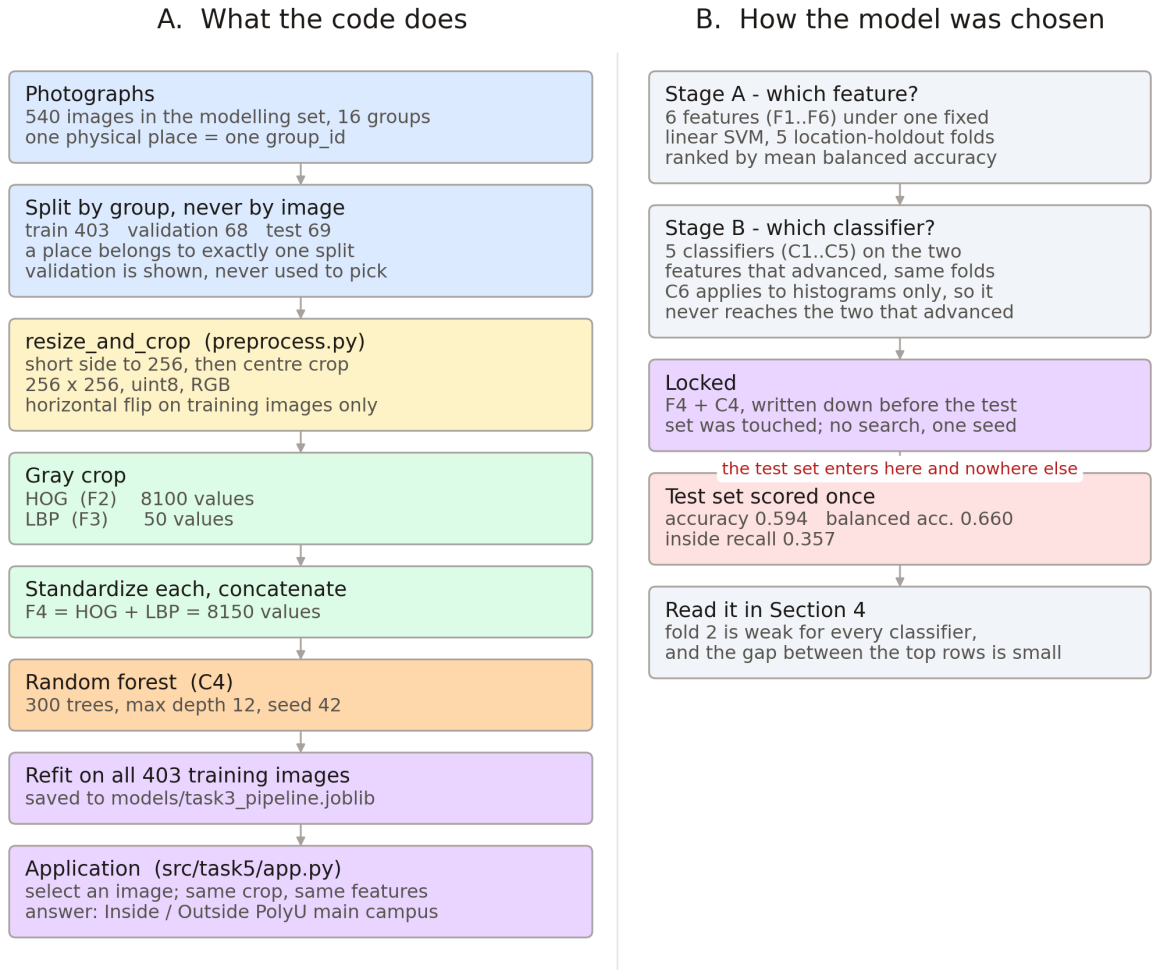


Figure 1. The pipeline, and the order in which the model was chosen. Left: what the code does, from the photographs through the split, the crop, the two features and the random forest to the saved pipeline that the application loads. Right: how the model was chosen. The comparison runs on the training places alone, in two stages, and the test set is reached once, after the choice has been written down.

Two things about the figure are worth stating in words. The first is that the left-hand column is a single path: the photograph goes through one resizer, one pair of features and one classifier, and the object that comes out is the file the application loads. There is no second path, and the application cannot reach the label any other way, because the label is produced by `predict_rgb` and by nothing else. The second is that the right-hand column is ordered. The feature comparison and the classifier comparison both run on the training places alone, in two stages, and only after both of them have finished is the test set read, once. The red line in the figure marks that boundary, and everything about how the result in Section 4 should be read follows from it.

Deep networks and CNN features are not used. The classifier reads pixels only.

1.4 Local application

Section 6 is the desktop program that loads `models/task3_pipeline.joblib` and shows one photograph's prediction. Sections 4 and 5 are the validation report and the single test evaluation of that same object. The application does not choose features and does not rescore the test set.

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2. Dataset

2.1 Places and the collection strategy

One physical place is one location group. Nearby views of the same courtyard, building, or landmark stay together. Separate buildings in the same district do not. HKU's modern teaching blocks and the Main Building are different groups. West Kowloon's M+ museum, Art Park, and harbourfront promenade are different groups. Continuous PolyU walkways stay in one group.

Inside the group, variation is recorded in columns rather than new groups: `sub_area` for the part of the place, and `scene_class` for one of 13 scene types. The four shooting axes also stay inside the group. Making any one of them its own group would turn the split into a split on that axis.

Axis	How it was varied
Viewpoint	About six stations. Facing direction (front, 45 degrees, side), camera height (ground, slight upward, downward from steps), and horizontal placement of the subject (left, center, right).
Distance	Two stations, each with a near, middle, and far frame.
Close view	Wall--ground junctions and materials, with spatial depth kept in the frame. Isolated objects were not collected.
Lighting	Night frames kept in the daytime group of the same place, so day and night of one place cannot fall into different splits.
Background	At least three background kinds per group. The same kind is used on both sides of the label when possible.

The collection quotas were within-group targets, not the final counts. Final counts are below.

Both classes contain buildings, covered walkways, green space, and open paving. Eleven of the thirteen scene types appear on both sides of the label. The two types that appear only outside are deliberately placed there: the five indoor frames of the HKU corridor, so that ``indoors'' cannot become a synonym for inside, and the nine market-stall frames, six of them on Temple Street and three on the Jordan Road stretch, where a night market would otherwise be a free cue.

Among the 718 usable photographs, before the per-group cap:

Scene type	inside	outside
Building façade	63	79
Covered walkway	73	63
Urban distance view	24	54
Tree-lined path	26	48
Green space	15	35
Ramp or steps	34	18
Building entrance	28	9
Construction hoarding	14	5
Paved plaza	14	16
Road frontage	10	57
Water	5	14
Indoor space	0	5
Market stall	0	9

Row sums equal 718. Façades and covered walkways are the two largest types (142 and 136). Water has 19 photographs. Road frontage is unbalanced (10 inside, 57 outside). Construction hoarding is temporary (14 inside, 5 outside). These limits are discussed in Section 2.6.

Night frames are 31/306 = 10.1% of usable inside images and 56/412 = 13.6% of usable outside images. The rates are close, so brightness alone is a weak class cue. All 31 campus night frames stay in the daytime group of the same place (IN-01, IN-02, IN-04, or IN-05).

No.	Place	Usable images
01	Walkway, J block to VA block	71
02	Coffered walkway, P/Q block	62
03	University Library and central podium	25
04	FSCH, VS block and swimming pool	53
05	Inner lawn and clock tower, main entrance	38
06	Building courtyard and Block A entrance	40
07	CMA courtyard	17
	total, before the per-group cap	**306**

The counts are the usable images with status include. They sum to 306, which is the usable inside total before the 50-image cap and before class balancing, so the modeling set of Section 3 holds fewer. The map keeps the building letters of the official campus map, so a place can be found on the ground from the figures alone.

2.2 Hard negatives

Red is the easiest shortcut. Much of the main campus is red brick, and a color histogram can score well by voting ``red means inside''. Outside photographs were chosen to block specific inside appearances:



Figure 2. The seven inside places on a PolyU campus map, one numbered pin per place. The pin numbers are the ones in the table below. Marker 05 is on the south side of J Core; the place it marks is the inner lawn and clock tower at the main entrance. The gate threshold itself is in the excluded set, not in IN-05.

Outside group	Inside appearance it matches
OUT-06 HKU Main Building (test)	Red-brick façades at IN-01, IN-03, and IN-04. Edwardian red brick and granite. Paired with IN-03, which is also in the test set.
OUT-02 HKU modern blocks and links (train)	Coffered walkway IN-02 and covered links IN-06. Includes the five indoor corridors.
OUT-09 West Kowloon harbourfront (val)	Entrance steps IN-05 and the large stair at IN-06.
OUT-08 Art Park (train)	Open lawns and courts at IN-02, IN-06, and IN-07. Bound to the same split as OUT-05 because the two sites see each other across a road.
OUT-05 M+ (train)	Modern buildings and hard plazas.
OUT-04 Elements (train)	Large roofs over open space, compared with the coffered ceiling at IN-02.

Outside group	Inside appearance it matches
OUT-03 Kowloon Park (train)	Lawns, courtyards, and water. Eleven red-tile Chinese-garden frames are an extra red-brick negative.
OUT-07 HKU campus green (test)	Lawns at IN-05 and the courtyard green at IN-07. Not a red-brick negative. Paired with IN-07 in the test set.
OUT-01 Jordan (train)	Dense street façades, as a contrast to campus blocks.

2.3 Labels, exclusions, and anchors

734 photographs were annotated. 16 were excluded. 718 remained, in 16 places (7 inside, 9 outside). After a cap of 50 images per group and class balancing, the modeling set has 540 images, 270 inside and 270 outside.

Excluded	Count	Reason
IN-05	11	Name board at the gate (4) or covered turnstile passage (7). Notes on the manifest rows match this split.
OUT-01	2	A full license plate in the foreground. `OUT-01_020_IMG_0816` reads `WX 9304` on the photograph (checked 2026-10-15). `OUT-01_052_IMG_1003` carries `VA 7003` in the manifest notes; that frame was not re-read in the same pass.
OUT-02	2	Street signs on the outer side of the campus boundary.
OUT-03	1	Several recognizable faces.

Another 22 frames had been queued for a plate or face check. Each was looked at and returned to the usable set. That decision was not taken from a model score.

Phone transfer stripped EXIF GPS. Ground truth anchors are signs that are visible in the frame, including the university library sign and crest (IN-03 `IMG_1459`), Stanley Ho Building (IN-04 `IMG_1229`), CMA Building and Lui Che Woo Building (IN-07 `IMG_1428`), ``THE UNIVERSITY OF HONG KONG'' on the Main Building (OUT-06 `IMG_0863`), and Jordan Road (OUT-01 `IMG_1002`). Every anchor is a sign that is visible in the frame, so each label can be re-checked from the photograph alone. The campus name board `IMG_1304` is excluded and is not used as an anchor.

2.4 Why this scale is enough, and where it is not

There is no required image count. The unit that has to be learned and tested is the place, not the frame. Sixteen places are enough to put several places of each class in training, hold out paired places for a five-fold location check, and still leave a validation pair and a test pair. Adding more frames inside a place that is already in training does not add an unseen place.

The cap and the balance step exist so that one large place cannot dominate the pixel count. OUT-01 entered with 82 usable frames and is stored as 50. The smallest place, OUT-07, has only 7 frames. That group is why the split cannot sit on 60/20/20, as the next section shows. Sixteen places are enough to run the protocol. They are not enough to make the test estimate sharp: the test set can hold only four places.

2.5 Split, leakage controls, and counts

`data/annotations/splits.csv` is the split. It was not redrawn for this report. `src/tools/build_dataset.py` refuses to finish if a group crosses splits, if OUT-05 and OUT-08 are separated, or if any split is missing a class. Augmentation is horizontal flip, applied only to training images after the split. Subsampling inside a capped group is evenly spaced so the viewpoint range remains.

Image counts and groups, from `data/annotations/dataset_summary.md`:

Split	Class	Images	Groups	Group ids
train	outside	213	6	OUT-01, OUT-02, OUT-03, OUT-04, OUT-05, OUT-08
train	inside	190	4	IN-01, IN-02, IN-04, IN-06
val	outside	30	1	OUT-09
val	inside	38	1	IN-05
test	outside	27	2	OUT-06, OUT-07
test	inside	42	2	IN-03, IN-07

Totals: train 403, validation 68, test 69, sum 540. By images that is 403/540, 68/540, and 69/540 (about 74.6%, 12.6%, and 12.8%). By groups it is 10/16, 2/16, and 4/16, which is 62.5%, 12.5%, and 25%. The target is a split of location groups aiming at 60/20/20, with both classes in every split. Both classes are present in every split. The group shares are the ones to read. OUT-07 has 7 images and cannot fill a 20% outside test group by itself; the test outside side is OUT-06 plus OUT-07, and the validation outside side is the single group OUT-09.

The validation split is the one place where the 60/20/20 target is not met, at 2 of 16 groups, and the reason is worth stating plainly. There are 7 inside groups and 9 outside groups, and a validation split that holds out one group per class costs one group from each side. Holding out two groups per class instead would leave 5 inside and 7 outside groups for training, but it would also take IN-05, the main entrance, out of training. IN-05 is the group that covers the

boundary case, and a model that never sees a threshold view in training has no way to learn that a threshold is not a scene. The alternative was rejected for that reason, and the cost is recorded here rather than argued away: the validation score rests on one place per class and is therefore reported, and not used, as the selection score.

Stored counts per place (original usable → kept):

Group	Place	Class	Usable	Kept	Split
IN-01	Walkway, J Core to VA	inside	71	50	train
IN-02	P/Q coffered walkway	inside	62	50	train
IN-03	University Library and central podium	inside	25	25	test
IN-04	FSCH / VS / pool	inside	53	50	train
IN-05	Main entrance, inner lawn and clock tower	inside	38	38	val
IN-06	Block court and Core A entrance	inside	40	40	train
IN-07	CMA Building courtyard	inside	17	17	test
OUT-01	Jordan	outside	82	50	train
OUT-02	HKU modern blocks and links	outside	57	50	train
OUT-03	Kowloon Park	outside	72	32	train
OUT-04	Elements	outside	62	30	train
OUT-05	West Kowloon, M+	outside	21	21	train
OUT-06	HKU Main Building	outside	20	20	test
OUT-07	HKU campus green	outside	7	7	test

Group	Place	Class	Usable	Kept	Split
OUT-08	West Kowloon, Art Park	outside	38	30	train
OUT-09	West Kowloon har- bourfront	outside	53	30	val

Place names in this table are the full names. The generated summary in `data/annotations/dataset_summary.md` uses shorter labels for a few outside rows; the group id, class, counts, and split are the summary's.

2.6 Dataset limits that affect later numbers

- The test estimate rests on four places. A single difficult building moves the accuracy.
- OUT-07 contributes only 7 test images.
- Road frontage is 10 inside versus 57 outside in the usable pool. A model could treat heavy traffic as outside. The four test places are a library, a courtyard, a brick building, and a lawn, so Section 5 does not attribute the test misses to that imbalance.
- Indoor evidence exists only on the outside class, and only as 5 frames.
- Construction hoarding will disappear. It is a durability limit, not a stable campus cue.
- Night shares (10.1% and 13.6% of the usable pool) sit a little above a 5--8% target. They are similar across classes.
- IN-01 (train) and IN-05 (val) both include views around the Innovation Tower and VA. Validation accuracy is expected to look stronger than a fully unseen place. Model choice does not use it.

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3. Method

3.1 Design

The comparison was fixed before the run: the six features, the classifiers, the five folds, the single selection score, and the tie-break rules were all written down first.

Stage A asks which representation helps. Six hand-crafted features share one classifier: standardization, where the classifier needs it, and a linear SVM with `C = 1`. Stage B asks whether the classifier matters. The two features with the highest location-holdout score are each trained with the classifiers in Section 3.4. Parameters are fixed. There is no search on the validation set and none on the test set.

The selection score of a configuration is the mean balanced accuracy of five location-holdout folds. Balanced accuracy is the average of the outside recall and the inside recall, so a fold that only recognizes one class does not look successful. Validation accuracy is written down and is not part of the sort. The sort key in `src/task3/cv.py` is that mean balanced accuracy; it was read back from the code and checked against the published tables.

Ties on features keep the smaller dimension, then the earlier id. Ties on classifiers keep the simpler one in this order: logistic regression, linear SVM, multinomial naïve Bayes, k-NN, random forest, histogram gradient boosting. If the winner had been a linear SVM, one extra row would

have reduced that feature to 100 principal components, fit on the training fold only, and trained an RBF SVM. The winner was not a linear SVM, so that row was not added.

3.2 Preprocessing

`resize_and_crop` is the only resizer.

1. Open the file as RGB with Pillow. OpenCV's BGR reader is not used, so a later interface cannot silently disagree with evaluation.
2. Scale the short side to 256 with area interpolation, keeping the aspect ratio.
3. Center-crop to 256×256. A landscape frame loses equal strips on the left and right. A portrait frame loses equal strips on the top and bottom.
4. Return `uint8` of shape 256×256×3. HOG, LBP, and SIFT convert that crop to gray. The color histogram uses HSV of the same crop.

Horizontal flip is a training augmentation only. Validation and test images are not flipped. Color jitter is not used. Night and lighting were collected on purpose; jitter would hide that axis.

The random seed is 42.

3.3 Features

Id	Feature	Setting	Dimension
F1	HSV histogram	H 16 × S 16 × V 8. Whole frame plus a 2×2 grid. Each histogram L1-normalized, then concatenated.	10240
F2	HOG	9 orientations, 16×16 cells, 2×2 blocks, L2-Hys, on gray [1].	8100
F3	LBP	Uniform pattern, P = 8, R = 1, 10 bins [2]. Whole frame plus 2×2, each block L1-normalized.	50
F4	HOG + LBP	F2 and F3 standardized separately on the training images, then concatenated.	8150

Id	Feature	Setting	Dimension
F5	Bag of words	SIFT [4] keypoints, at most 500, in a bag-of-words histogram [5]. Mini-batch k-means, 256 words, fit on that fold's training descriptors only. Word counts L1-normalized. An image with no keypoints becomes a zero histogram and is counted in the log.	256
F6	Bag of words + spatial pyramid	F5 on the whole frame, plus four histograms on a 2×2 grid [6].	1280

SIFT is `cv2.SIFT_create` from `opencv-contrib` [9]. If that function is missing, the run stops. ORB is not a substitute. In the environment used for the 2026-10-14 run, SIFT was available.

The vocabulary is not stored across folds. A fold that is building a codebook never sees the held-out place.

What each of the six measures is set out in Section 11.1.

3.4 Classifiers

Id	Classifier	Setting
C1	Linear SVM	`LinearSVC`, `C = 1`, standardized features
C2	Logistic regression	`C = 1`, `lbfgs`, standardized features
C3	k-NN	k = 5. Cosine distance on histogram features. Euclidean distance when the feature includes HOG.
C4	Random forest	300 trees, `max_depth = 12`, `random_state = 42` [3]
C5	Histogram gradient boosting	`max_depth = 6`, `random_state = 42`
C6	Multinomial naïve Bayes	`alpha = 1`, only for non-negative L1 histograms (F1, F3, F5, F6)

C1, C4, and C5 are the three main families (SVM, random forest, gradient boosting). C2 and C3 are different decision rules. C6 is a simple histogram control. F2 and F4 skip C6 because they are not non-negative histograms.

C6 was additionally evaluated on the four non-negative L1 histogram features (F1, F3, F5, F6) as a supplementary comparison. Its rows are in `outputs/task3/stage_a_nb.csv`. They do not enter the feature ranking, which is computed from the C1 rows of `stage_a.csv` alone, so the locked pipeline is unaffected.

Feature	C1 linear SVM	C6 naive Bayes	C6 inside recall	C6 outside recall
F1 colour histogram	0.538	0.499	0.177	0.822
F3 LBP	0.588	0.500	0.000	1.000
F5 bag of words	0.566	0.504	0.008	1.000
F6 bag of words + spatial pyramid	0.563	0.572	0.345	0.800

What each of the six classifiers does is set out in Section 11.2.

The result is a useful one. Naive Bayes collapses on this data: on F3 and F5 its inside recall is 0.000 and 0.008, so it answers outside for almost every campus frame and its balanced accuracy settles at 0.500, the score of a constant predictor. Its best result is 0.572 on F6, still far below the 0.721 that F4 reaches under the linear SVM, and above the 0.563 that the same feature reaches under it. The ranking of the features and the locked pipeline are therefore unchanged whichever of the two protocols is applied: F4 with the linear SVM is the strongest configuration in both tables. What the comparison adds is a second, independent instance of the behaviour the test set exposes at the end of this report, a classifier that has learned the shape of the outside class and nothing else.

3.5 Location holdout

Each training place is a single class, so a fold always holds out one inside place and one outside place together. OUT-05 and OUT-08 are one atom. The five folds are written in `src/task3/config.py`:

Fold	Held-out inside	Held-out outside
0	IN-01	OUT-01
1	IN-02	OUT-02
2	IN-04	OUT-03
3	IN-06	OUT-04
4	IN-01	OUT-05 and OUT-08

IN-01 is held out twice because there are five outside atoms and four inside training groups. The five balanced accuracies are averaged as they are. The two IN-01 folds are not merged.

On each fold the feature model (codebook, standardization, and PCA if any) is fit on the other training images, including their horizontal flips. Flips are not placed in the held-out set. The held-out places are scored with balanced accuracy and with both recalls.

3.6 Locked pipeline and how to rerun it

After Stage B, the winning pair is refit on all training images plus horizontal flips. It is saved with joblib. Validation is scored once for the report. The test set is scored once. Per-image test predictions were written to `outputs/task3/test_predictions.csv`, which ships with the code archive, so every test number below can be recomputed from it.

Commands, from the repository root. The environment is Python 3.12 with the pins in `requirements.txt`. The path of the virtual environment is specific to the machine that trained the model and is not printed here.

```
python -m src.task3.run_stage_a
python -m src.task3.run_stage_b
python -m src.task3.evaluate
```

Pinned packages in `requirements.txt`:

```
numpy==2.4.3
Pillow==12.1.1
opencv-contrib-python==5.0.0.93
scikit-image==0.26.0
scikit-learn==1.8.0
joblib==1.5.3
```

`opencv-contrib-python` is required. The non-contrib wheel has no SIFT, and the feature module exits rather than switching descriptors.

The crop cache key includes the image size (`s256__filename`). `run_stage_a` and `run_stage_b` hash `splits.csv` and `src/task3/*.py`, write that fingerprint to a stamp file beside the results, and refuse to reuse a results table when the fingerprint does not match or the stamp is absent, unless `--force` is passed. The two commands were run again on 2026-10-23 from the code in the archive: both reproduced `outputs/task3/stage_a.csv` and `outputs/task3/stage_b.csv` byte for byte, with the same feature ranking, the same locked F4 with the random forest, and the same per-fold columns, so the stamp beside the results now matches the shipped source and a rerun resumes instead of prompting for `--force`. The test evaluation in Section 4.4 was not rerun; it remains the single evaluation of 2026-10-14. The raw result tables and their provenance note, `outputs/task3/RESULTS_PROVENANCE.txt`, ship with the code archive, so every number in Section 4 can also be checked without rerunning anything.

Code layout: `src/task3/config.py`, `data.py`, `preprocess.py`, `features.py`, `models.py`, `cv.py`, `run_stage_a.py`, `run_stage_b.py`, `evaluate.py`. Dataset construction is `src/tools/build_dataset.py`. File names are resolved only in `data.py`, to find a path. `transform` receives arrays. The local window is `src/task5/app.py` and `src/task5/predict.py` (Section 6). It loads the saved pipeline and does not fit a new one.

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4. Experimental results

Selection uses location-holdout balanced accuracy only. The validation column is display. The figures in this section are drawn by `docs/figures/plot_task4_locked.py` from the locked counts, and every number in the tables below is the value in the corresponding file under `outputs/task3/`.

The comparison is a set of configurations rather than an exhaustive search. Stage one fixes the classifier at the linear SVM so that the feature axis is isolated; stage two varies the classifier only on the two features that advanced. Every classifier was run at one fixed setting, with no parameter search anywhere in the protocol.

4.1 Stage A: linear SVM, six features

Feature	Location-holdout BA	Inside recall	Outside recall	Validation accuracy
F4 HOG + LBP	0.721	0.698	0.744	0.779
F2 HOG	0.708	0.677	0.740	0.750
F3 LBP	0.588	0.536	0.641	0.779
F5 bag of words	0.566	0.453	0.679	0.824
F6 bag of words + pyramid	0.563	0.414	0.712	0.809
F1 color histogram	0.538	0.495	0.581	0.544

F4 and F2 enter Stage B.

The per-fold balanced accuracies behind those means are the `ba_0` to `ba_4` columns of `outputs/task3/stage_a.csv`, folded as in Section 3.5.

Feature	fold 0	fold 1	fold 2	fold 3	fold 4	mean
F1 HSV histogram	0.570	0.390	0.542	0.654	0.532	0.538
F2 HOG	0.820	0.760	0.514	0.696	0.753	0.708
F3 LBP	0.650	0.490	0.462	0.667	0.673	0.588
F4 HOG + LBP	0.840	0.770	0.534	0.708	0.752	0.721
F5 bag of words	0.580	0.610	0.528	0.579	0.533	0.566
F6 bag of words + pyramid	0.500	0.640	0.512	0.592	0.573	0.563

The color histogram is last. That matches the reason HKU was collected: red brick is on both sides of the label, and color does not survive a change of place. Bag-of-words has the highest

validation accuracies (0.824 and 0.809) and location-holdout scores near 0.56. The validation pair is the entrance lawn against the West Kowloon harbourfront, which is visually easy, and a codebook can memorize keypoints of buildings it has seen. Validation accuracy is the wrong column to maximize. It is shown so that this disagreement stays visible.

Adding the spatial pyramid on top of bag of words does not help: F6 scores 0.563 against F5's 0.566, so the finer spatial layout buys nothing at this vocabulary size.

4.2 Stage B: F4 and F2 across classifiers

C6 does not apply to F2 or F4. The winner is not a linear SVM, so the RBF row was not run.

Feature	Clf	Location-holdout BA	Inside recall	Outside recall	Validation accuracy
F4	C4 random forest	**0.757**	0.660	0.855	0.691
F2	C4 random forest	0.748	0.649	0.846	0.676
F2	C3 k-NN	0.740	0.562	0.918	0.691
F4	C3 k-NN	0.739	0.556	0.922	0.706
F2	C2 logistic regression	0.725	0.685	0.765	0.735
F4	C2 logistic regression	0.722	0.701	0.744	0.765
F4	C5 histogram gradient boosting	0.721	0.648	0.793	0.603
F4	C1 linear SVM	0.721	0.698	0.744	0.779
F2	C1 linear SVM	0.708	0.677	0.740	0.750
F2	C5 histogram gradient boosting	0.701	0.626	0.775	0.676

The locked pipeline is **F4 HOG+LBP + C4 random forest**.

On the same F4 features, the linear SVM's validation accuracy is 0.779 and the forest's is 0.691. The forest is still the one selected, because its location-holdout balanced accuracy is higher (0.757 against 0.721). k-NN's outside recall is higher (0.922 on F4) and its inside recall is 0.556. On fold 2, which holds out IN-04 and OUT-03, k-NN labels almost all of the held-out campus as outside. The forest's balanced accuracy on that fold is 0.616, the highest of the five F4 classifiers.

The five folds of the location holdout, balanced accuracy for the five F4 classifiers, are the evidence behind that mean. They are the folds of Section 3.5: fold 0 holds out IN-01 and OUT-01, fold 1 IN-02 and OUT-02, fold 2 IN-04 and OUT-03, fold 3 IN-06 and OUT-04, and fold 4 IN-01 together with OUT-05 and OUT-08.

F4 classifier	fold 0	fold 1	fold 2	fold 3	fold 4	mean
C1 linear SVM	0.840	0.770	0.534	0.708	0.752	0.721

F4 classifier	fold 0	fold 1	fold 2	fold 3	fold 4	mean
C2 logistic regression	0.840	0.780	0.534	0.696	0.763	0.722
C3 k-NN, k=5	0.840	0.700	0.510	0.783	0.861	0.739
C4 random forest	0.840	0.790	**0.616**	0.700	0.841	**0.757**
C5 histogram gradient boosting	0.870	0.700	0.561	0.750	0.722	0.721

The forest does not win every fold. It takes fold 1 at 0.790 and fold 2 at 0.616, is second on fold 4 at 0.841 behind k-NN at 0.861, and on fold 0 it shares 0.840 with three other classifiers behind gradient boosting at 0.870, the best single fold in the table. On fold 3 it comes fourth at 0.700. k-NN has the opposite shape: it wins fold 3 and fold 4 but is last on fold 2 at 0.510, the same collapse the validation column hides, and on fold 0 it is tied for the lowest. What the mean rewards is the middle of the distribution rather than its best point, and the forest has the highest floor of the five: its worst fold is the 0.616 of fold 2, and every other classifier drops below that somewhere, k-NN to 0.510, gradient boosting to 0.561, and both linear models to 0.534. These are the `ba_0` to `ba_4` columns of `stage_b.csv`, copied without rounding beyond three decimals.

The gradient boosting fold profile is uneven: it is the best of the five on fold 0 at 0.870, second on folds 2 and 3, and last on folds 1 and 4. That is why its mean sits with the linear models rather than with the forest.

The same five folds on the second feature, from the same file, give the same evidence for F2:

Feature	Clf	fold 0	fold 1	fold 2	fold 3	fold 4	mean
F2	C1 linear SVM	0.820	0.760	0.514	0.696	0.753	0.708
F2	C2 logistic regression	0.830	0.770	0.529	0.712	0.782	0.725
F2	C3 k-NN, k=5	0.860	0.730	0.530	0.758	0.822	0.740
F2	C4 random forest	0.820	0.770	0.575	0.713	0.861	0.748
F2	C5 histogram gradient boosting	0.770	0.720	0.575	0.675	0.763	0.701

The forest wins on F2 as well, 0.748 against 0.740 for k-NN, and the weak fold is the same one: on fold 2, which holds out IN-04 and OUT-03, every F2 classifier falls between 0.514 and 0.575. Changing place, not changing classifier, is what costs the score.

F4 logistic regression has a higher holdout inside recall than the forest (0.701 against 0.660) and a lower balanced accuracy (0.722 against 0.757). The written score is balanced accuracy. The test set's later inside recall does not reopen that choice.

F4 gradient boosting and F4 linear SVM both print as 0.721, which is a rounding artefact and not a tie. The unrounded means in `stage_b.csv` are 0.72060 for gradient boosting and 0.72089 for the linear SVM, so the SVM is the marginally stronger of the two. Neither reaches the forest at 0.757, so the winner is unchanged.

The representation moves the score more than the classifier does. From color to HOG+LBP under the same linear SVM, location-holdout balanced accuracy goes from 0.538 to 0.721. On F4, the five classifiers span about 0.721 to 0.757.

One seed was used, 42, and the folds are not independent: IN-01 is held out in two of the five folds, so there are effectively four independent inside training places. The fold-to-fold spread is far wider than the gaps between the top features or between the top classifiers. F4 leads F2 by 0.013 balanced accuracy and the forest leads the linear SVM on F4 by 0.036, while the selected configuration alone moves from 0.616 on fold 2 to 0.841 on fold 4. Small differences between configurations should not be read as a real ordering.

4.3 Validation after the refit

The forest was refit on the full training split, with horizontal flips, and then applied to the 68 validation images (IN-05 38, OUT-09 30).

Accuracy **0.691**. Balanced accuracy **0.710**.

	Predicted outside	Predicted inside
True outside	26	4
True inside	17	21

Class	Precision	Recall	F1	Support
outside	0.605	0.867	0.712	30
inside	0.840	0.553	0.667	38

These three-decimal figures are the published rounding of the matrix: 47/68 = 0.691, outside recall 26/30 rounds to 0.867, inside recall 21/38 rounds to 0.553. Seventeen campus frames are called outside, and four harbourfront frames are called inside. Inside recall is already 0.553 on this easier pair. Figure 3 draws the same four cells, and each cell also carries its share of the true-class row. Section 2.6 is why 0.691 is not quoted below as generalization.

4.4 Test, one evaluation

69 images: IN-03 25, IN-07 17, OUT-06 20, OUT-07 7.

Accuracy **0.594**. Balanced accuracy **0.660**.

	Predicted outside	Predicted inside
True outside	26	1
True inside	27	15

	Predicted outside	Predicted inside

Class	Precision	Recall	F1	Support
outside	0.491	0.963	0.650	27
inside	0.938	0.357	0.517	42

Working from the same cells: accuracy 41/69 rounds to 0.594; outside precision 26/53 rounds to 0.491; outside recall 26/27 rounds to 0.963; outside F1 is 0.650; inside precision 15/16 rounds to 0.938; inside recall 15/42 rounds to 0.357; inside F1 rounds to 0.517; balanced accuracy rounds to 0.660. Figure 4 draws the same four cells the same way. The same matrix recomputes from `outputs/task3/test_predictions.csv` (outside 0.4906 / 0.9630 / 0.6500, inside 0.9375 / 0.3571 / 0.5172).

Two constant classifiers frame this result, and one of them is better on accuracy.

Predictor	accuracy	balanced accuracy	outside recall	inside recall
Always outside	27/69 = 0.391	0.500	1.000	0.000
Always inside	42/69 = 0.609	0.500	0.000	1.000
Locked pipeline	41/69 = **0.594**	**0.660**	0.963	0.357

The test split holds 42 inside images and 27 outside ones, because the split unit is the place and the two held-out inside places are larger than the two held-out outside places. A constant that says inside therefore scores 0.609, slightly above the model's 0.594, so on accuracy alone this model does not beat that constant. Accuracy is the wrong headline for a split with that composition. On balanced accuracy, which weights the two classes equally, the model is 0.160 above either constant. It keeps 0.963 outside recall where the always-inside constant has 0.000, and when it does say inside it is right 15 times out of 16. The error is missed campus scenes, not a flood of false campus predictions.

Two comparisons have to be kept apart, and the weaker one should not be used to carry the argument.

Against the always-outside constant the model is clearly ahead. It recognises 15 inside images that the constant calls outside, and gives up 1 outside image in exchange, a net gain of 14 correct answers, and it is the better detector of the inside class.

Against the always-inside constant the model is behind on accuracy, and nothing here changes that number. The always-inside constant is not a candidate for the application, because a predictor that calls every photograph campus has no outside class to detect and cannot answer the question the program exists to answer. The model does answer it, on 26 of the 27 outside photographs.

One point belongs in the other direction, because it is a weakness of the evidence rather than of the model. The selection score was fixed before the run and it was balanced accuracy, not

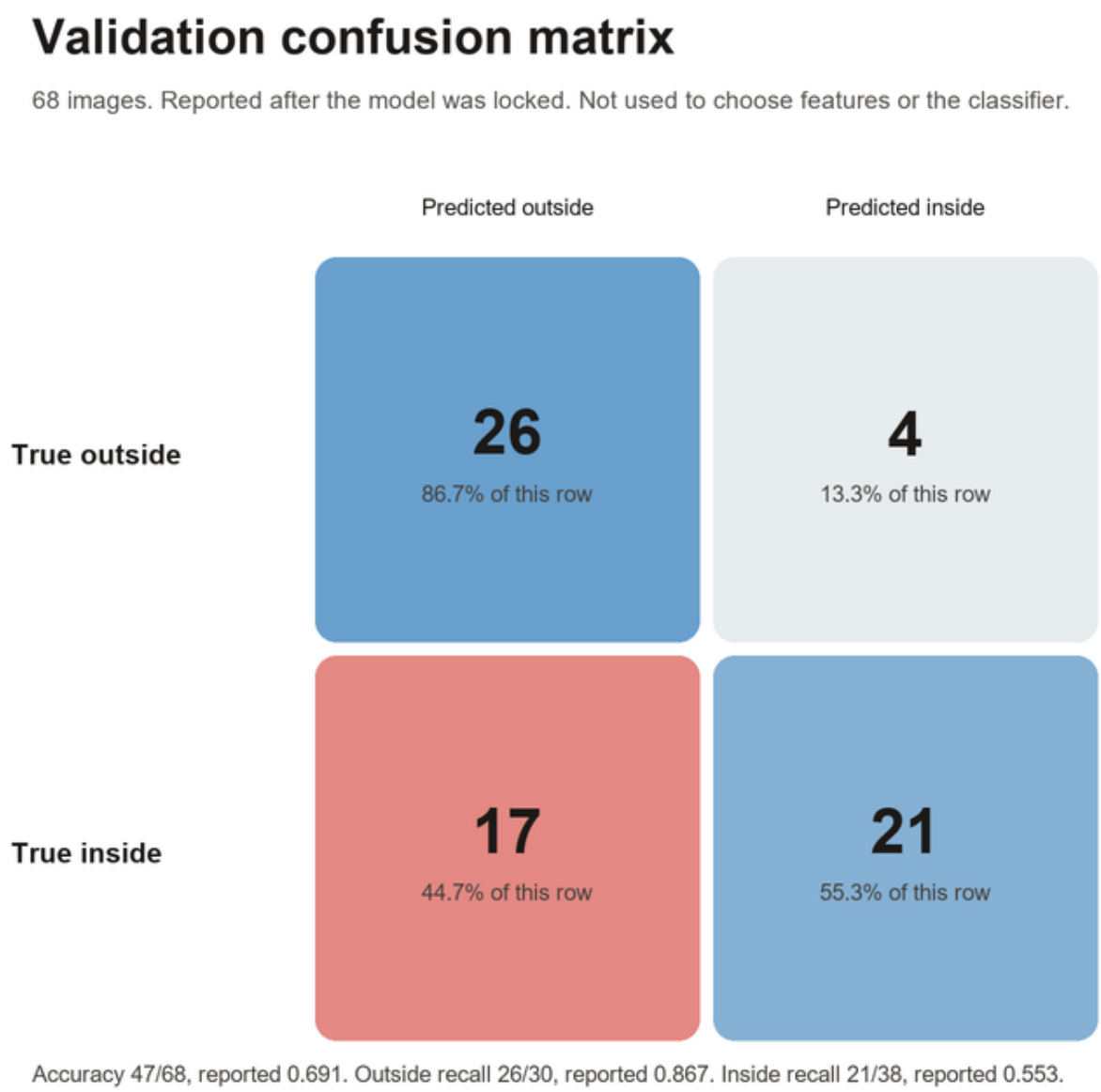


Figure 3. Validation confusion matrix of the locked pipeline. Counts from `outputs/task3/val_metrics.json`. Not used to choose the model.

accuracy. On this test composition the two constants sit at 0.391 and 0.609 and the model sits at 0.594, so the metric that was chosen before the data was seen happens to favour the model on this split, while the model is behind the stronger constant on the metric that was not chosen. The gap between the two constants is 0.218 and the gap between the model and the always-inside constant is 0.015, so the ranking at the top of the table rests on a small margin, as Section 4.2 shows. The two errors are also not symmetric for an application: 1 outside photograph is called campus, and 27 campus photographs are called outside. Which of those a user would rather have is a product decision, not a measurement, and the report does not pretend to have made it.

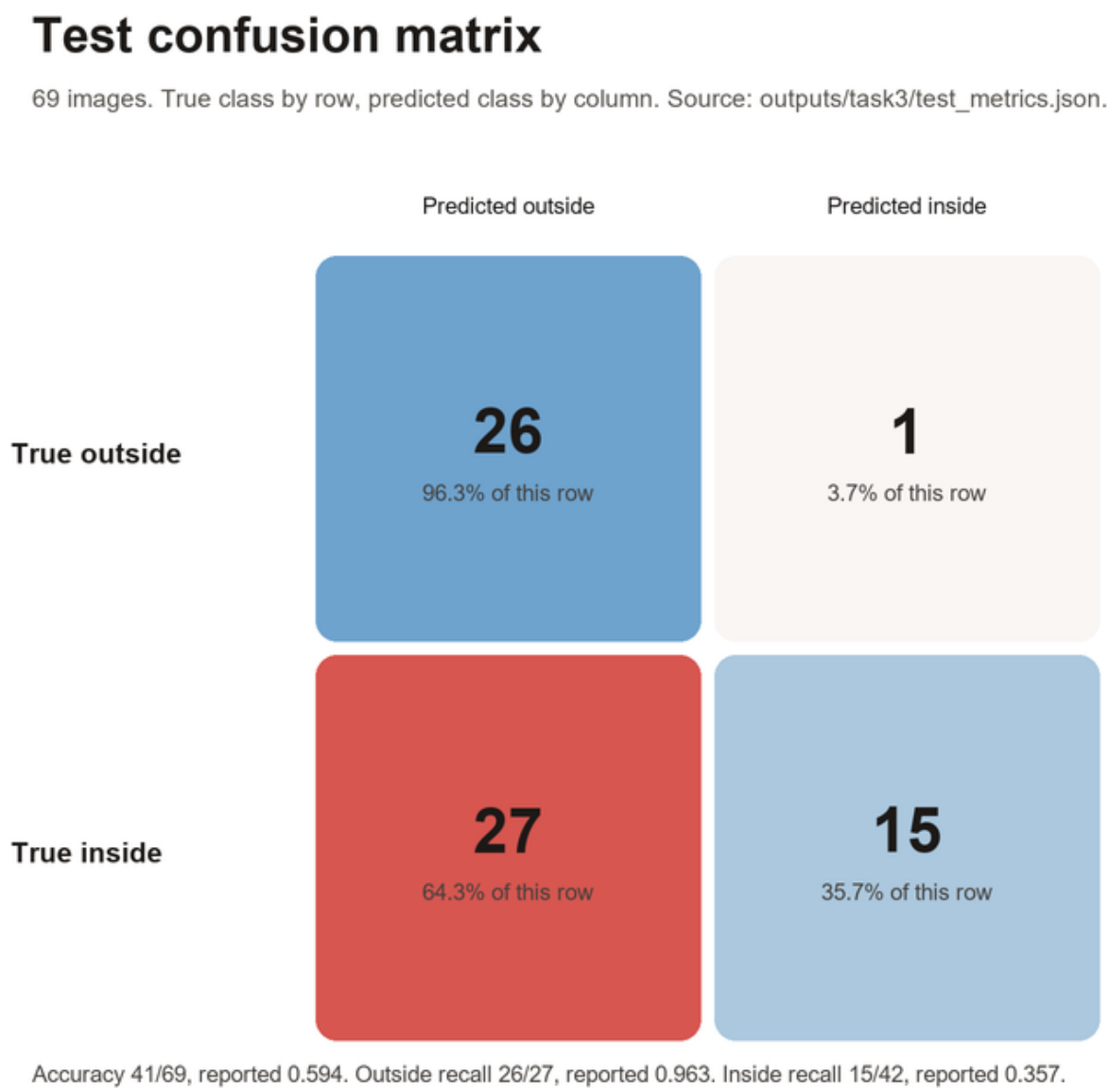


Figure 4. Test confusion matrix. Same locked pipeline, evaluated once.

5. Error analysis

This section reads the test matrix place by place, and then reads the frames themselves.

5.1 How to read the test matrix

The four cells are not one story.

- 26 of 27 outside images are correct.
- 1 outside image is called inside. That is the only false inside in the test set.
- 27 of 42 inside images are called outside. This cell is the failure.
- 15 inside images are correct.

Outside precision is 0.491 because 27 campus misses sit in the same predicted-outside column as the 26 true outside images. Inside precision is 0.938 because the inside column is almost clean. Accuracy 0.594 hides that split. Balanced accuracy 0.660 still sits far below the location-

holdout score of the same configuration (0.757), which was measured on other places and on other training subsets.

5.2 Failure by place

The test predictions were split by group. Each group has one label, so a group's errors can land in only one off-diagonal cell.

Group	Correct / n	Accuracy	Where the errors go
OUT-06 HKU Main Building	19/20	0.950	1 image predicted inside. This is the only false inside in the matrix.
OUT-07 HKU campus green	7/7	1.000	None.
IN-03 Library and central podium	9/25	0.360	16 images predicted outside.
IN-07 CMA courtyard	6/17	0.353	11 images predicted outside.

Check: 19+7+9+6 = 41, the numerator of 41/69. The 16 and 11 misses sum to the 27 false outsides. The single OUT-06 miss is the single false inside. OUT-07 cannot contain that false inside. Figure 5 plots these four accuracies against the overall test accuracy.

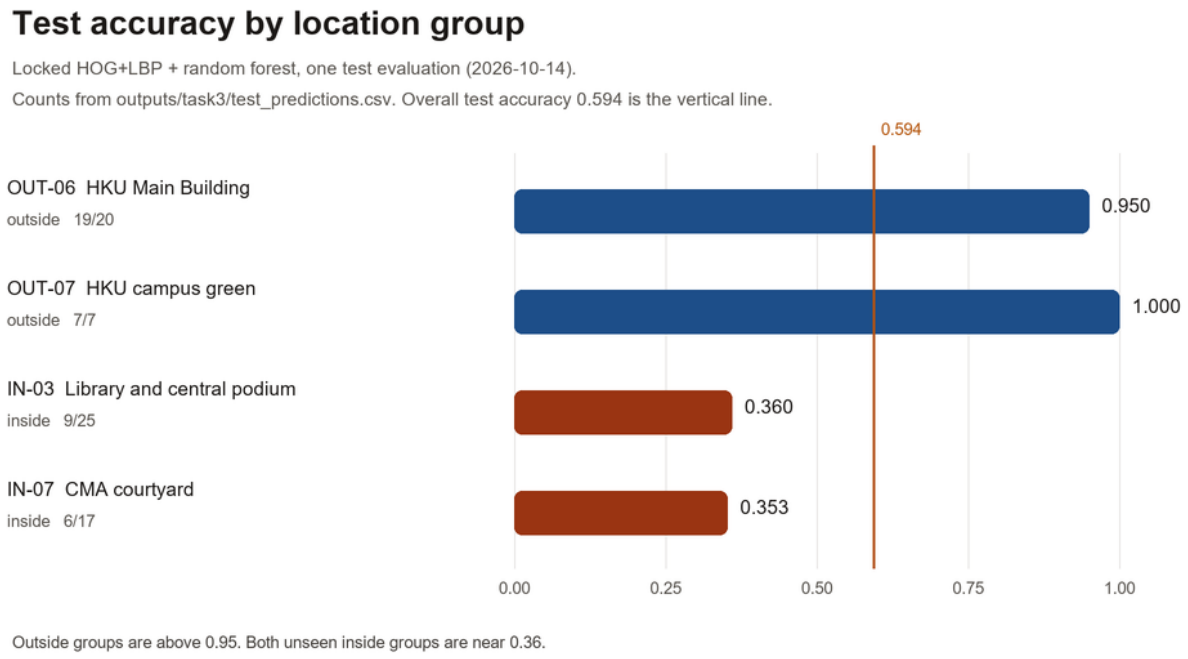


Figure 5. Test accuracy of the four held-out places. The vertical line is the overall test accuracy 0.594.

The pattern is stable across the two unseen campus places, and it is stable across the two unseen outside places in the opposite direction:

1. Unseen campus places are mostly predicted outside (0.360 and 0.353).

2. Unseen outside places are almost all predicted outside (0.950 and 1.000).
3. The red-brick hard negative works. HKU Main Building is not accepted as PolyU.
4. The library, which is also red brick and was paired with OUT-06 for that reason, is mostly predicted outside as well (16 of 25). The model does not vote ``red brick means inside''. It pushes both brick sites toward outside, and the campus site is the one that suffers.
5. Green space does not transfer either. OUT-07 is 7/7 outside. IN-07, the courtyard collected as the visual partner, is 6/17.

On the final refit, inside recall is 0.553 on validation and 0.357 on test. During selection, the same feature and forest had a mean inside recall of 0.660 across the five training-location folds. Those three numbers use different fits. They are not epochs of one curve. They do show the same direction: campus scenes that were not in the fit are the ones dropped.

On the unseen campus places the model behaves like an approximate constant predictor toward outside. It is approximate because 15 campus images are still recovered. Calling it a literal constant would erase that cell.

5.3 Relation to HOG and LBP

F4 is 8100 HOG dimensions plus 50 LBP dimensions (`src/task3/config.py`). Under a linear SVM, adding LBP moves location-holdout balanced accuracy from 0.708 to 0.721. Under the forest, it moves from 0.748 to 0.757. The texture block is a small add-on. The test failure belongs to this local gradient representation, with LBP not supplying the missing cue.

HOG counts edge orientation inside 16×16 cells. It does not encode which building this is, or which side of the campus boundary the camera stands on. Uniform LBP at radius 1 is a local texture histogram. Color is discarded. Most of the building's layout is discarded. After concatenation, a forest of depth 12 can split training places whose edge patterns it has seen. The training campus places are the J--VA walkway, the P/Q coffered walkway, FSCH / the pool, and the Core A court. The test campus places are the library podium and the CMA courtyard. Those edge statistics do not have to match.

Two representations that might seem more semantic were measured on the location holdout and were worse:

- Color is the cue most tempted by red brick. It scored 0.538, last in Stage A. It was not evaluated again on the test set, so no test figure for it exists. The test behavior of the locked model is still informative: OUT-06 is rejected (19/20) while IN-03 is also mostly rejected (16 of 25 predicted outside). Removing color removed an unreliable shortcut. It did not leave a feature that can say ``this brick building is the library''.
- Bag-of-words looked best on validation and failed the change of place (0.566 and 0.563). A codebook of training keypoints is a memory of training buildings. That is why it lost the selection, and why a high validation number was not allowed to overrule the holdout.

The training counts, 213 outside and 190 inside, are too close to explain 53 outside predictions out of 69. The bias shows up in the recalls of the selected forest itself (holdout means 0.855 outside and 0.660 inside) and becomes sharp on the four test places. The mechanism consistent with the feature definition is that a new campus layout falls into regions of the HOG space that the forest already associates with outside scenes.

Four explanations are worth separating before the misses are attributed to anything: background bias, lighting, similar places, and viewpoint.

Candidate explanation	What is measured here
Similar-looking places	Measured. OUT-06 against IN-03 (brick), OUT-07 against IN-07 (green / court). Outside is recognized. Inside is about one third correct.
Background bias	The scene table was built so that 11 of 13 types exist in both classes. Test errors cluster by place. That does not support a story in which one background type produces all 27 misses.
Lighting	Night shares on the 718 usable images are 10.1% inside and 13.6% outside. No color jitter was applied. The prediction table carries no day/night field, and none of the 69 test images comes from a `night/` folder, so the test set contains no night frame at all. A night-error rate is reported as not applicable rather than estimated.
Viewpoint	Viewpoints vary inside each place by design, and the error table has no viewpoint column, so no viewpoint-error rate is computed. The qualitative set does show that distance is not what separates the correct CMA frames from the missed ones: a close view of the patterned wall is correct and a wider view of the same wall is not, and a wide courtyard view is correct while another wide courtyard view is not. What the crop contains, rather than how far away the camera stood, is what Section 5.4 reads frame by frame.
Road-frontage imbalance	10 inside versus 57 outside in the usable pool. The test places are not that road set. The misses are not attributed to traffic.

This is the ceiling of HOG+LBP on a place it has not seen, under a protocol that refuses to train on that place. The number is final for this pipeline; it is not a grid that stopped halfway.

5.4 Qualitative examples

Figure 6 is the contact sheet of individual test predictions, and it follows this discussion. The collection map is Figure 2 in Section 2.1.

The sheet was produced by `docs/figures/plot_task4_locked.py` from `outputs/task3/test_predictions.csv` and the photographs in `data/pictures/`, the directory the dataset archive from `DATASET_LINK.txt` unpacks into. The script refuses to draw unless the prediction table still sums to 26, 1, 27, 15 and the four place rates still equal 19/20, 7/7, 9/25 and 6/17, so a figure can only ever be drawn from the locked predictions. `docs/figures/task4_example_manifest.csv` records which file sits in which cell.

The thirteen frames were chosen by a rule fixed in advance, not by how they look: within each cell, in filename order --- 2 correct OUT-06, 2 correct OUT-07, the single false inside (necessarily OUT-06), 2 missed IN-03, 2 missed IN-07, 2 correct IN-03, and 2 correct IN-07. A frame was never picked because it appeared to support the conclusion.

The grid shows the two failure modes at once. The one false inside is `OUT-06_012_IMG_0892.jpg`, a glass entrance and a flight of steps at HKU, not the red-brick facade that dominates its group --- the 10 OUT-06 facade frames are all correct.

The missed inside frames are `IN-03_001_IMG_1444.jpg` and `IN-03_002_IMG_1445.jpg`, where the brick rib wall of the library fills the frame, and two frames at the CMA courtyard, `IN-07_005_IMG_1431.jpg` and `IN-07_006_IMG_1432.jpg`. Nine further CMA frames are missed as well, but the sampling rule takes the first two in filename order, so only those two are in the grid. The two correct CMA frames are `IN-07_002_IMG_1428.jpg` and `IN-07_001_IMG_1427.jpg`.

The crops are what the model sees, and no single cue separates the four CMA frames. What can be said is what each crop contains. The two correct CMA frames keep the identifying mark: `IN-07_002` has the patterned screen with the PolyU crest centred under open sky, and `IN-07_001` has the crest small on the far left with the planted courtyard and the white canopy below it. The two missed ones do not. In `IN-07_005` the crest is reduced to a sliver at the right edge, the frame holds two patterned screens and the brick building above a reflecting pool, and in `IN-07_006` the crest has left the frame altogether, with a tree, café parasols and the brick building taking the crop instead.

The library frames miss for the opposite reason. `IN-03_001` and `IN-03_002` are the café frontage under the brick ribs, with the LibCafé sign at the bottom of the frame, while `IN-03_003` and `IN-03_004` are the library's own entrance, with the 大學圖書館 sign, the glass doors and a lawn, and both of those are classified correctly. The proportion of brick is not the discriminator here: the correctly classified library frames carry more brick than the two that are missed. What the misses share is that the identifying feature of the place is absent from the crop or replaced by something adjacent, a crest at the frame edge, no crest at all, a café rather than the library's entrance.

That is the limit of the representation rather than of the collection. HOG and LBP describe local gradient and texture, so a crop made of brick ribs, paving and café furniture is a texture field like any other, and there is no channel in which ``this particular crest'' or ``this particular sign'' could be encoded. The material cue is also the one the design deliberately taught the model to distrust by putting Hong Kong University's brick into the outside class: OUT-06, a brick building, is 95% correct, and a campus frame that shows mostly brick is the same texture as that negative. It is not a viewpoint problem and not a class-balance problem --- the same entrance at two scales gives one correct answer and one wrong one --- it is that the representation has one strong material cue and no identity cue.

An earlier, smaller grid of examples was drawn from the same prediction table and is kept in `docs/figures/task4_examples.png`. It names eight photographs, five of which --- `OUT-06_001_IMG_0863.jpg`, `OUT-06_012_IMG_0892.jpg`, `OUT-07_001_IMG_0934.jpg`, `IN-03_001_IMG_1444.jpg` and `IN-07_002_IMG_1428.jpg` --- also appear in Figure 6; the other three are not, and eight of the thirteen frames in Figure 6 appear in no other grid. The two grids are not the same set. Where they overlap they agree on the outcome of each frame, because both were read from the same prediction table.

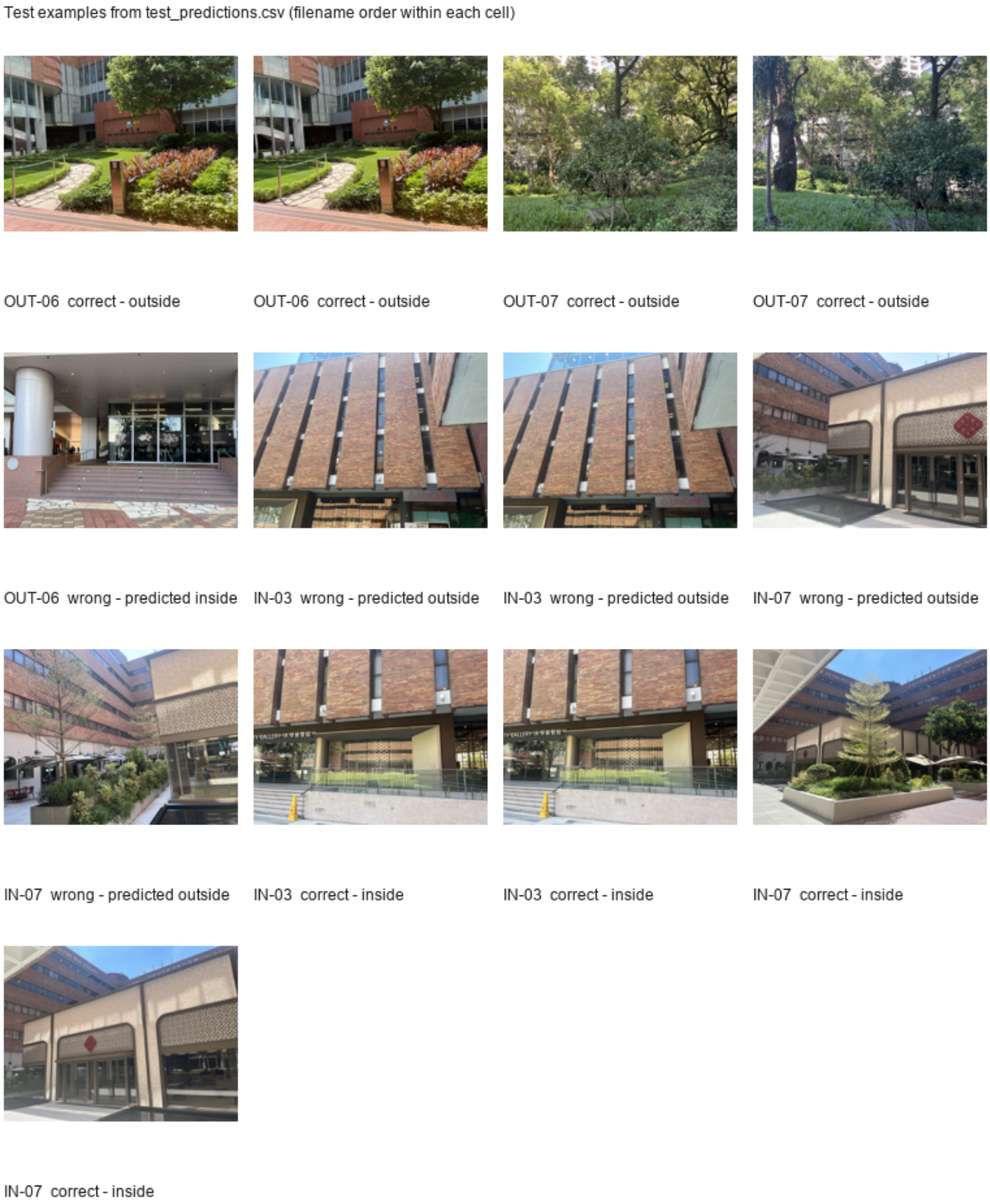


Figure 6. Thirteen test photographs in the order of the manifest: each caption gives the location group and whether the locked pipeline classified that frame correctly. The file name of every frame is in `docs/figures/task4_example_manifest.csv`.

5.5 The test numbers do not retune the model

Accuracy 0.594 and inside recall 0.357 are the score of the pipeline that was locked on location-holdout balanced accuracy. IN-03 and IN-07 stay in the test set. Features are not re-ranked on these 69 images. The forest is not swapped for logistic regression because logistic regression had a higher holdout inside recall. A later check of whether more campus places would change the picture needs new places and a new held-out evaluation of this same pipeline. It does not move the current test groups into training.

---

6. Application

The local program lets a user select one scene photograph, shows that photograph, and returns either ``Inside PolyU main campus'' or ``Outside PolyU main campus''. It is a desktop window written with Tk from the Python standard library. Nothing was added to `requirements.txt`. Pillow, joblib, OpenCV [9], scikit-image [8] and scikit-learn [7] are the training dependencies. The checks are listed in Section 6.4. The same launch lines are in the root `README.md`.

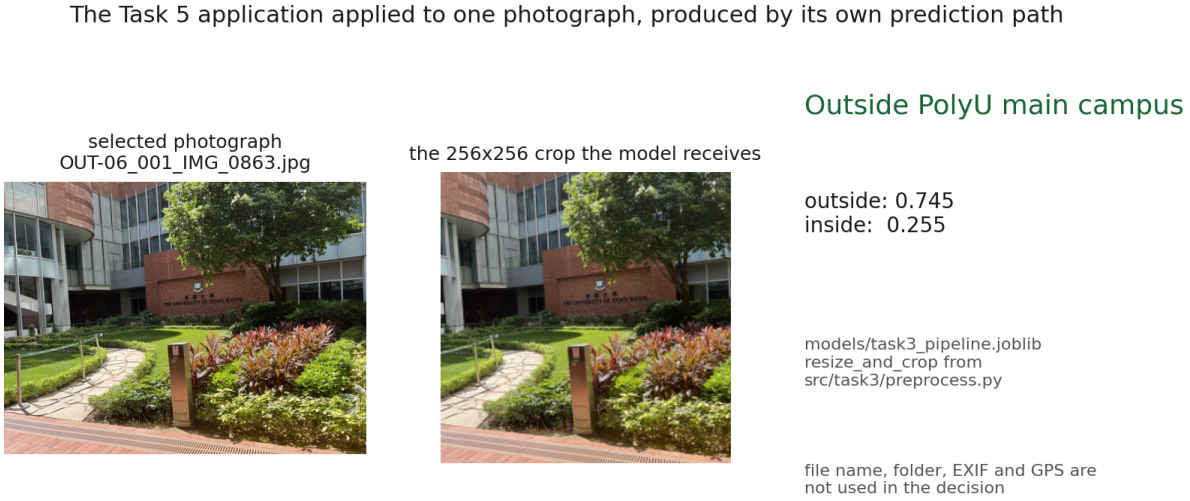


Figure 7. The application applied to one photograph, drawn by the program's own prediction path. The left panel is the selected file, the middle panel is the 256×256 centre crop the model receives, and the right panel is what the program returned: the sentence and the two class probabilities. The last two lines of that panel are the figure's own note, that the file name, folder, EXIF and GPS are not used, not output from the program. The photograph is `OUT-06_001_IMG_0863.jpg`, the Hong Kong University Main Building, one of the hard negatives, and the locked pipeline calls it outside with 0.745.

The program does not train and does not edit `splits.csv`. The classifier remains the locked HOG+LBP random forest in `models/task3_pipeline.joblib`. The test accuracy 0.594 and the other Section 4--5 figures are unchanged.

6.1 How to launch

From the repository root, in the same environment as the training code:

```
python -m src.task5.app
```

That opens the window. With no display, the same loader and the same preprocessing print one image and do not open a window:

```
python -m src.task5.app --image path\to\photo.jpg
```

The command prints the English label and, for this forest, the outside and inside probabilities. If the file cannot be read, it prints the error and exits with status 1.

On Windows, `pythonw -m src.task5.app` opens the window without a console. The official Windows installer includes Tk. If Ubuntu reports that `_tkinter` is missing, install `python3-tk`. If `models/task3_pipeline.joblib` is absent, the window tells the user to run `python -m src.task3.evaluate` and does not run that training step itself. In this submission the saved file is already present.

6.2 What the window shows

The window title is ``PolyU Main Campus Classifier''.

1. **Select image...** opens a file dialog. Ctrl+O does the same. The dialog accepts jpg, jpeg, png, bmp, tif, tiff, and webp.
2. The left pane shows the selected photograph and its original width and height. The file name is written next to the button so the user can see which file is open. That string is not given to the classifier.
3. The right pane prints the sentence stored on the pipeline as `display_`: ``Inside PolyU main campus'' or ``Outside PolyU main campus''.
4. This random forest implements `predict_proba`, so the window also draws the inside probability and the outside probability.
5. The thumbnail is the array returned by `resize_and_crop`: the 256×256 center crop that the model receives. The widget scales that crop for display. The array passed to `predict` stays 256×256×3.

A line at the bottom of the window names `models/task3_pipeline.joblib` and states that the file name, folder, EXIF, and GPS are not used. Figure 7 shows the window with one photograph loaded: the file, the crop the model receives, and the two probabilities it returned.

6.3 How prediction is called

`src/task5/predict.py` is the prediction path. `src/task5/app.py` is the window and the `--image` printer. The order matches `src/task3/evaluate.py`: crop, then predict.

1. `load_rgb` in `src/task3/preprocess.py` reads the file as RGB `uint8`. It does not rotate from EXIF. Training and evaluation did not rotate, so the application does not add a rotation the model never saw.
2. `resize_and_crop` in the same module scales the short side to 256 and center-crops to 256×256×3. There is no second resizer.
3. `pipeline.predict([cropped])` returns 0 for outside or 1 for inside. The argument is a one-image list. A bare array would be iterated by rows.
4. `pipeline.predict_proba([cropped])` fills the two probabilities. If a future estimator had no `predict_proba`, the window would say so and would still show the label. This forest has the method.
5. `pipeline.display_[label_id]` is the English sentence written when the pipeline was saved: `{0: "Outside PolyU main campus", 1: "Inside PolyU main campus"}`.

`joblib.load` returns a scikit-learn [7] `Pipeline` with two steps. `features` is `ImageFeatures(feature_id="F4")`, which is `HogLbp`. `clf` is `RandomForestClassifier` with 300 trees, `max_depth=12`, and `random_state=42`. `classes_` is `[0, 1]`.

`HogLbp.transform` calls `resize_and_crop` again. When the image is already 256×256, that function returns the same array. The call written in the application is still `src.task3.preprocess.resize_and_crop`.

`predict_rgb` accepts an RGB array. The path is used only to read those pixels and to label the button row with the file name.

6.4 Checks recorded for the application

These checks were run on the installed `requirements.txt`. They do not refit the forest and they do not replace Section 4 or Section 5.

- From open to label, one image takes about 0.1 seconds, including HOG+LBP, `predict`, and `predict_proba`.
- Every crop is `(256, 256, 3)`. Calling `resize_and_crop` again on that crop leaves the array unchanged. Predicting the uncropped original returns the same label as cropping first.
- The same pixels saved as `IN-99_inside_campus.jpg` and as `OUT-99_outside_campus.jpg` produce the same label and the same probabilities. The name in the window changes. The sentence does not.
- An unreadable text file opens an error dialog and keeps the previous label. The command-line form exits with status 1.
- Noise, a 1×1 image, a very wide image, all black, all white, a grayscale JPEG, a PNG with an alpha channel, and a 3840×2400 JPEG each returned one of the two English sentences.
- The window opened at about 880×585. After a file was chosen, the canvas showed the original, the thumbnail showed the crop, and the prediction text was one of the two sentences. `python -m src.task5.app --image <file>` and `python src/task5/app.py --image <file>` printed that same label and both probabilities.

All 69 test images were replayed through the program's own prediction path, `predict_rgb` in `src/task5/predict.py`, which is the function the window calls. The label it returned matched the locked prediction table on 69 of 69 images. That is the direct check that the application and the evaluation are one pipeline: the same `models/task3_pipeline.joblib`, the same `resize_and_crop`, the same pixels, and the same answer. The replay is reproduced by running `python -m src.task5.app --image <file>` over the 69 files listed in `outputs/task3/test_predictions.csv` and comparing the labels.

Two scenes from the course material, a courtyard and a street, each 960×1080 pixels, were also run through the program to confirm that it returns a label and does not crash on a photograph it was not trained on. They are not part of the test set, and they were not used to change the model.

Example scene	Pixels	Program output	outside	inside
Courtyard	960×1080	Outside PolyU main campus	0.512	0.488
Street	960×1080	Inside PolyU main campus	0.460	0.540

Both probability pairs sit near one half. That is this locked pipeline on two scenes it was not trained on. It is not a new test accuracy. The held-out test result remains accuracy 0.594, inside recall 0.357, and the four place rates in Section 5.

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7. Discussion

Six questions about how the program was designed and tested, how far its robustness reaches, what went wrong along the way, how AI was used, and where that use stops. The answers use the dataset, the result tables under `outputs/task3/`, and the application checks in Section 6.4.

How do you design and test the program?

The geographic rules came first: main campus versus not, boundary frames dropped, one physical place as the split unit. Collection then filled both classes with matched scene types and with outside places that look like campus, especially HKU brick.

The learning program was designed as two locked stages rather than a search over every feature--classifier pair. Stage A compares six pixel features with one linear SVM. Stage B compares classifiers on the two winning features. The score is the mean balanced accuracy of five prewritten location-holdout folds. Validation accuracy is printed and ignored by the sort. The test set is applied once, after the pipeline is refit on all training images and saved.

Testing of the data side is the check inside `build_dataset.py`: no group crosses a split, the two mutually visible West Kowloon groups stay together, and every split contains both classes. Testing of the numbers is a separate read-only verification pass over the result files: the published tables were matched against `outputs/task3/stage_a.csv` and `outputs/task3/stage_b.csv` value for value, and the test confusion matrix was recomputed from `outputs/task3/test_predictions.csv`. That pass did not re-run the three training commands, so its statements about feature dimensions are a reading of the code rather than a measurement, and its reading of a licence plate was wrong and had to be re-checked against the photograph. The commands and package pins are in Section 3.6.

The desktop program is tested apart from that selection. One array saved under `IN-99_inside_campus.jpg` and under `OUT-99_outside_campus.jpg` yields the same label and the same probabilities; a second `resize_and_crop` on a 256×256 crop does not change the array; an unreadable file keeps the previous window label and makes `--image` exit with status 1. Those checks are about the interface. The replay of the 69 test images is in Section 6.4.

How do you ensure the robustness of the program in a real scenario?

Robustness here means a new place, a renamed file, a change of lighting, and a lookalike building. Three of those four are measured rather than argued, and the one that is not, a change of lighting, is the one the test set cannot cover: it holds no night frame at all, as Section 5.3 records.

- Lookalike buildings are in the data. OUT-06 is the brick negative. The color histogram, which would have used that shortcut, scored worst on the location holdout (0.538) and was not selected.
- Eleven of thirteen scene types occur in both classes, so a background such as ``walkway'' or ``lawn'' is not privately owned by one label.
- Night frames stay in the place's own group. The training code does not jitter color, so lighting remains a real axis instead of being augmented away.

- The split unit is the place. Bursts, revisits, day and night, and edits of one place stay in one split. Flips happen only after the split, and only on training images.
- The classifier never sees the file name. Renaming a photograph cannot flip the prediction except by changing pixels. Section 6.4 records that check: the same pixels under an `IN-` name and an `OUT-` name keep the same label and the same probabilities.
- Evaluation and the desktop window both call `resize_and_crop` and load `models/task3_pipeline.joblib`, so the reported test score is the score of the object the window loads.
- The cache key includes the crop size, and a rerun stops when the split file or the training code has changed. That prevents a silent mix of old crops and new code.
- The crop keeps the short side and centre-crops, so a portrait frame loses strips at the top and the bottom. The two example scenes in Section 6.4 are 960×1080 portrait frames, and both were classified wrongly. A portrait campus scene is a known weak case here, not an untested one.
- The prediction was replayed over the whole test split through the program's own path, and returned the locked label on 69 of 69 images. Robustness of the pipeline is therefore a measured property of the object the application loads, not an argument about two code paths that look similar.

The measured limit is in Section 5. On two campus places the model had not seen, inside recall is 0.357, and 53 of 69 predictions are outside. A real ``is this the main campus?'' decision that must not miss campus scenes is not supported by that recall. More unseen campus places would be required before trusting the negative prediction. The present test groups stay held out.

What problems do you find and how do you solve them?

Problem	What was done
GPS disappeared after the phones were copied to a computer.	Labels were anchored to signs visible in the frame (Section 2.3). GPS was never a feature.
Gate photographs are ambiguous and some show people.	11 IN-05 frames were excluded, with notes on the manifest.
Two Jordan frames show a readable plate; one Kowloon Park frame shows faces.	Those frames were excluded. The plate on `OUT-01_020_IMG_0816` was corrected to `WX 9304` after looking at the image.
Red brick can act as an inside shortcut.	HKU Main Building and part of Kowloon Park were collected as red outside scenes. The color histogram then scored last on the location holdout, which is consistent with that design.
The validation pair is easy, and IN-05 overlaps the Innovation Tower / VA views already in IN-01.	Validation is reported and is not the selection score.
Bag-of-words memorizes buildings and wins on validation (0.824) while losing the holdout (0.566).	The codebook is fit inside the fold, and the holdout score is the one that ranks features. Bag-of-words was not selected.
k-NN's outside recall looks strong while its inside recall falls to 0.556.	Balanced accuracy is the score. The forest wins it without that collapse.

Problem	What was done
A crop cache keyed only by file name would reuse a 256×256 image after a size change.	The key now includes the size. This was fixed after the published run and was not a reason to rerun selection.
All seven inside manifests once shared one place name, which reads like a leaked group.	Each inside group now has its own place name. The 540-way split of filenames, groups, and labels was unchanged.
Test inside recall is 0.357.	Section 5 gives the breakdown. The pipeline was not retuned on the test set. The remedy is a new evaluation rather than a new fit: more inside places under the same boundary rules, one further held-out group, and the saved pipeline scored on it, as set out in Section 8.
A name such as `IN-99` could be read as the class.	`predict_rgb` receives the RGB array only. The same pixels under `IN-99_inside_campus.jpg` and `OUT-99_outside_campus.jpg` return the same label and the same probabilities (Section 6.4).

How do you use GenAI to assist you in the implementation?

I used AI for what it is good at: writing code quickly and running experiments quickly.

I started with information search. I read about how scene photographs are classified without deep learning, and collected a list of candidate representations: colour histogram, HOG, LBP, SIFT bag of words, spatial pyramid, together with the classifiers usually paired with them.

I then had the AI build the comparison code for that list. It wrote the feature extractors, the classifier wrappers, the five-fold location-holdout loop that refits the codebook and the scaler inside each fold, the dataset scripts that build the split table and check it, and the scripts that draw the figures. Each stage runs from one command.

What I decided myself: the protocol, written down before any result existed (five folds fixed in advance, one selection score, no tuning, the test set scored once), the place labels, which frames to keep or drop, and the photographs themselves whenever a generated note and the picture disagreed.

The experiments then ran, and the choice followed the numbers rather than my expectation. The pair with the best location-holdout score won; a feature with the best validation accuracy was dropped because it did not survive a change of place.

After the selection I improved that model: the refit on all training images, the saved pipeline, and the desktop program that loads it, with the test set kept out of every decision.

The files the model drafted are `src/tools/build_dataset.py`, `src/tools/collapse_to_venues.py`, `src/tools/apply_naming.py`, `src/tools/make_contact_sheet.py`, the modules under `src/task3/`, and `docs/figures/figure1_inside_locations_annotated.png`. They did not take the photographs, assign places, or decide include versus exclude; those decisions were made by looking at the frames.

Verification was reading that material back against the data, and it is named here so that it can be repeated. A read-only pass confirmed that the sort key in `src/task3/cv.py` is the mean location-holdout balanced accuracy, that the published Stage A and Stage B tables match `outputs/task3/stage_a.csv` and `stage_b.csv` value for value, and that the test confusion matrix recomputes from `outputs/task3/test_predictions.csv`. Two of those checks appear above with the numbers attached: the five folds of the location holdout in Section 4.2, and the resolved tie between F4 gradient boosting and the F4 linear SVM in the same section. The 69-image replay in Section 6.4 is the third, and the only one run after the selection was locked.

How does GenAI understand and solve the tasks?

The generated code treats the work as a reproducible classical pipeline: parse a split file, crop, extract a feature, fit a scikit-learn classifier, write metrics. That part is sound, and it is the path that produced the Stage A and Stage B tables.

Its picture of the dataset was less reliable, and the errors show where the general recipe replaced the local rule:

- After places had been collapsed from a finer draft to 7 inside groups, an earlier annotation of the campus map and Figure 2 still described 12 inside groups, including ids IN-09, IN-10, and IN-12 that are not in `splits.csv`. The figure and the hard-negative section were rewritten from the split file.
- The CMA building was first recorded as 香港中華煤氣. The sign in the photograph reads 香港中華廠商聯合會樓 (CMA Building). The manifest entry was corrected from the sign.
- An earlier exclusion annotation gave the plate on `OUT-01_020_IMG_0816` as `JF 0367`. A second visual pass suggested `UA 9364`. The photograph, read on 2026-10-15, shows `WX 9304` . The manifest now says `WX 9304`.
- Source manifest CSVs were left out of version control by a generated ignore rule, so only the derived `splits.csv` was tracked. The ignore rule was changed and the manifests were committed.
- When SIFT might be missing, the suggestion was to switch to ORB. The rule stated in Section 3.3 is to stop and revise the experiment, not to swap the descriptor quietly. The run kept SIFT because `cv2.SIFT_create` was present.

On the modeling choice, the generated tables correctly showed bag-of-words leading validation accuracy. Reading that column as the answer would have selected a feature that fails on a new place. The selection rule had already defined the answer as location-holdout balanced accuracy. The tables were kept, including the flattering validation column, and the holdout column was the one that ranked rows.

Twenty-two frames that a cautious pass had marked for plate or face review were put back to include only after each frame was looked at, not because a model accepted them.

What are the limitations and areas for improvement in GenAI's solution?

The useful limit is that a generated pipeline will complete a standard recipe and will be wrong on facts that live only in the photographs or in a rule written for this campus.

- The assistant read a licence plate wrong twice before a person read it correctly. The first annotation said `JF 0367`, a second visual pass said `UA 9364`, and the photograph read by hand says `WX 9304`. Two generated readings and one correct reading. The improvement is

to treat any fact about image content that the assistant writes down as a claim to be checked against the file, not against the assistant's second opinion.

- Place ids and building names drift when the list underneath them changes: an earlier annotation of the campus map and the generated map still described 12 inside groups after the split had been collapsed to 7. The improvement is to regenerate the map from the split file rather than from the map, which is what the rewrite did.
- A substitution of ORB for SIFT would have made the experiment irreproducible: the descriptor would have changed under the same name. The improvement is a hard stop, which is what the code does when SIFT is absent.
- A table of numbers invites a ranking, and validation accuracy was largest for the feature that does not travel. The safeguard adopted here is to freeze the selection column before the run and to publish the other column so the disagreement stays visible. That is what Section 4 does.
- The test set is scored once. Refitting on IN-03 or IN-07 to raise the 0.594 would break the holdout, and no such fit was made. That rule was fixed before the run rather than derived from the result.

The model-side limit in Section 5 is separate. It is a property of HOG+LBP on unseen places, measured on the locked pipeline. It is not repaired by another round of generation against the test score.

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8. Reflection

What I expected from a protocol fixed in advance was a ranking that would hold up, and a score that would carry to new places. The first expectation was only partly right, and the second was wrong.

The protocol did what it was built to do at the level of the rules: the color shortcut was rejected, the codebook that only looks good on the validation pair was rejected, and no classifier was chosen by validation accuracy. What the rules did not control was how much the score moves between folds. With one seed and effectively four independent inside training places, the fold-to-fold spread is wider than the gaps between the top configurations, so the order of the top rows is not something this much data can settle.

What the result taught is that the score carries less than the design assumed. The selected pair, HOG+LBP with a random forest at 0.757 location-holdout balanced accuracy, becomes accuracy 0.594 and inside recall 0.357 on the held-out test places, with OUT-06 at 0.950, OUT-07 at 1.000, IN-03 at 0.360, and IN-07 at 0.353. The outside places transferred almost perfectly and the campus places did not, which is the opposite of what a balanced accuracy measured across the training locations suggested.

Those test numbers are a usable measurement and a poor campus detector. The representation has no channel for ``this is PolyU'' once color is set aside and the building has not been seen. Local gradients generalize only as far as the next façade looks like a training façade. The library and the CMA courtyard do not. That limit belongs to the feature, and the test set was left untouched while it was found.

What would actually change the evidence is more inside places, collected under the same boundary rules, and one new held-out evaluation of this saved pipeline. The route is concrete: add inside places so that a fifth inside group can be held out, leave the current IN-03 and IN-07

groups in the test set and never train on them, and score the new group with the pipeline saved in `models/task3_pipeline.joblib` rather than with a refit.

Two measurements are worth making while that collection happens, and neither touches the locked model. The first is a colour-normalised variant of the same HOG and LBP, which would separate ``the crop was mostly brick and paving'' from ``the negative was never learnable''; Section 5.4 shows the misses clustering on crops in which the identifying mark of the place is missing, marginal or replaced by something adjacent, and only that experiment can settle whether the material is the problem. The second is a per-cell breakdown that also splits by scene class, because the aggregate table cannot show which part of a frame the model actually saw. Rewriting the current test score by training on it would not.

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9. References

1. N. Dalal and B. Triggs. Histograms of oriented gradients for human detection. In *IEEE Conference on Computer Vision and Pattern Recognition* (*CVPR*), 2005.
2. T. Ojala, M. Pietikäinen, and T. Mäenpää. Multiresolution gray-scale and rotation invariant texture classification with local binary patterns. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 24(7):971--987, 2002.
3. L. Breiman. Random forests. *Machine Learning*, 45(1):5--32, 2001.
4. D. G. Lowe. Distinctive image features from scale-invariant keypoints. *International Journal of Computer Vision*, 60(2):91--110, 2004.
5. G. Csurka, C. R. Dance, L. Fan, J. Willamowski, and C. Bray. Visual categorization with bags of keypoints. In *ECCV Workshop on Statistical Learning in Computer Vision*, 2004.
6. S. Lazebnik, C. Schmid, and J. Ponce. Beyond bags of features: Spatial pyramid matching for recognizing natural scene categories. In *CVPR*, 2006.
7. F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825--2830, 2011.
8. S. van der Walt, J. L. Schönberger, J. Nunez-Iglesias, F. Boulogne, J. D. Warner, N. Yager, E. Gouillart, T. Yu, and the scikit-image contributors. scikit-image: Image processing in Python. *PeerJ*, 2:e453, 2014.
9. G. Bradski. The OpenCV Library. *Dr. Dobb's Journal of Software Tools*, 2000.
10. The Hong Kong Polytechnic University. Campus map. `https://www.polyu.edu.hk/campus-map/`. Geographic reference for the inside class.

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10. Figure index

Figure	File	Role
1	`docs/figures/pipeline.png`	The pipeline and the order in which the model was chosen.
2	`docs/figures/figure1_inside_locations_annotated.png`	The seven inside places, one numbered pin per place. The numbers and counts are in the table in Section 2.1.

Figure	File	Role
3	`docs/figures/task4_val_confusion_matrix.png`	Validation confusion matrix, locked pipeline.
4	`docs/figures/task4_test_confusion_matrix.png`	Test confusion matrix, one evaluation.
5	`docs/figures/task4_test_per_group.png`	Test accuracy by place.
6	`docs/figures/task4_example_grid.png`	Thirteen test photographs, one per selected cell, labelled with the place and the outcome. The file list is in `docs/figures/task4_example_manifest.csv`.
7	`docs/figures/task5_application.png`	The desktop application applied to one photograph: the file, the 256×256 crop the model receives, and the label and probabilities it returned.

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11. Appendix: terms and abbreviations

The report uses short names for the feature sets, the classifiers and a number of standard ideas in image analysis. This section says what each one means here, so that a reader who has not worked with these particular methods can follow Sections 3 to 5.

11.1 The feature sets

A feature set turns one photograph into a fixed-length vector of numbers. `F1` to `F6` are the six that were compared; `F4` is the one the locked model uses.

Id	Short name	What it measures	Values
F1	HSV histogram	How much of each colour and brightness band the frame contains	10240
F2	HOG	How the brightness gradient points are oriented, in 9 direction bins, over 16×16 cells pooled in 2×2 blocks	8100
F3	LBP	A 10-bin histogram of the local binary pattern, each pixel compared with its 8 neighbours, on the whole frame and on a 2×2 grid	50
F4	HOG + LBP	F2 and F3, each standardized on the training images and then concatenated. **The locked feature set**	8150
F5	Bag of words	SIFT keypoints pooled into 256 visual words by k-means and counted	256
F6	Bag of words + spatial pyramid	F5 on the whole frame, plus four histograms on a 2×2 grid	1280

11.2 The classifiers

Id	Name	What it does here
C1	Linear SVM	A linear maximum-margin classifier. The fixed classifier of the feature comparison, so that stage one varies only the feature
C2	Logistic regression	A linear model whose output can be read as a probability
C3	k-NN	Classifies a photograph by the labels of its 5 nearest neighbours in feature space
C4	Random forest	300 decision trees voting on the answer. **The locked classifier**
C5	Histogram gradient boosting	Boosts small trees over histogram-binned features
C6	Multinomial naive Bayes	A count-based classifier. Valid only for non-negative histograms, so it applies to F1, F3, F5 and F6 and never reaches the two features that advanced

11.3 Features and preprocessing

Term	Meaning here
Augmentation	Mirroring the training photographs to enlarge the training set. Validation and test images are never mirrored
Centre crop	Cutting the middle square out of a scaled frame. A landscape frame loses equal strips on the left and the right; a portrait frame loses them at the top and the bottom
Gray	The single-channel version of the crop, used by HOG, LBP and SIFT
HSV	Hue, saturation and value. The colour space the colour histogram is computed in
L1 normalisation	Making a histogram sum to 1, so that it describes proportions rather than raw counts
L2-Hys	The HOG normalisation that also clips the largest gradients, so that one strong edge cannot dominate a cell
P and R	The local binary pattern's neighbourhood size and radius, here P = 8 and R = 1
Pipeline	The fitted scikit-learn object holding the feature extractor and the classifier together, so that training and the application run the same code
SIFT	Scale-Invariant Feature Transform, the local keypoint detector used by F5 and F6
Spatial pyramid	Splitting the frame into regions and histogramming each separately, so that where something appears matters and not only how much of it there is
Standardisation	Subtracting the mean and dividing by the standard deviation, using the training images only
uint8	Integers from 0 to 255, the pixel format the crop is returned in

11.4 Evaluation

Term	Meaning here
Balanced accuracy	The mean of the two class recalls. A constant predictor scores 0.500 on it, which is why it is the selection score
Confusion matrix	The 2×2 table of true class against predicted class
Fold	One repetition of the location holdout. There are five. Each holds out one inside place and at least one outside place; fold 4 holds out IN-01 together with OUT-05 and OUT-08, because those two outside places can be seen from each other
Hard negative	An outside-class photograph that looks like the inside class. Here the Hong Kong University buildings, collected so that the model cannot read brick as a campus cue
Inside recall	The share of campus photographs correctly called inside
Location holdout, also called venue holdout	A cross-validation in which every fold holds out whole places, so the model is always measured on places it has not seen
Precision	Of the photographs called inside, the share that really are inside
Recall	Of the photographs that really are inside, the share that are called inside
F1 score	The harmonic mean of precision and recall. Not to be confused with the feature set F1 in Section 11.1
Support	The number of test photographs in a class
Majority-class baseline	A predictor that always answers the more common class of the test set

11.5 The data and the split

Term	Meaning here
group_id	The identifier of a physical place. It is the unit that is split, so no place appears in two splits
Inside (1)	A photograph taken on the PolyU main campus, with the boundary taken from the official campus map
Outside (0)	A photograph taken anywhere else, including the campuses of other institutions
Seed	The fixed random seed, 42, used for the tree ensembles and the codebook. The five folds are not drawn at random: they are written out in `src/task3/config.py`
Split	One of train, validation or test. The validation split is reported but never used to choose anything